# Optimization Landscape Analysis of a Fixed Convolutional Sigmoid Network for Inverse Problems

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## Abstract

This work studies the optimization landscape of a fixed single-layer convolutional neural network used for inverse reconstruction. The model consists of a linear convolution operator followed by a sigmoid activation with controllable stiffness. Despite its simplicity, the resulting objective function exhibits non-trivial non-convex behavior. I analyze how the sigmoid stiffness parameter, convolution kernel, and target structure shape the geometry of the loss landscape and affect optimization performance. Through theoretical analysis and empirical evaluation, I show that increasing sigmoid stiffness leads to gradient sparsity and degraded optimization performance. Among tested methods, Adam demonstrates the highest robustness in challenging regimes. These findings highlight how activation nonlinearity and operator structure fundamentally alter optimization dynamics. All experiment code is available at: https://github.com/ahmedElmersawy/Optimization-Landscape-CNN.

## 1 Introduction

Understanding optimization landscapes is central to modern machine learning. While deep neural networks are often studied empirically, simplified models can provide valuable insights into fundamental optimization behavior.

In this work, I analyze a fixed single-layer convolutional neural network with a sigmoid activation. The goal is to solve an inverse problem: recovering an input that produces a desired binary output through the network. Despite its apparent simplicity, this problem exhibits rich non-convex behavior.

The analysis focuses on how three factors influence the optimization landscape: (i) the sigmoid stiffness parameter, (ii) the convolution operator, and (iii) the target structure.

Our key finding is that increasing sigmoid stiffness induces a transition from a smooth optimization regime to a sparse, boundary-dominated regime. In this regime, gradients vanish almost everywhere, effectively reducing the dimensionality of the optimization problem and forcing reconstruction to proceed through narrow transition regions corresponding to edges.

## 2 Related Work

Optimization in neural networks has been extensively studied, particularly in the context of non-convex loss landscapes (1). Prior work has examined how activation functions influence gradient flow, especially the saturation behavior of sigmoid nonlinearities (6).

Inverse problems in imaging often involve nonlinear forward models and are known to exhibit ill-posedness and non-uniqueness (2). Convolutional operators such as Sobel and Laplacian filters are

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widely used in signal processing, and their properties significantly affect reconstruction difficulty (5). The challenge of navigating ill-conditioned objectives motivates the use of adaptive methods (3), whose coordinate-wise rescaling is particularly suited to the anisotropic landscapes studied here.

## 3 Problem Setup

I consider the fixed single-layer convolutional neural network

$$f_\theta(x) = h(Ax), \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ is a convolution matrix corresponding to a fixed kernel $\theta$, and $h(\cdot)$ is an element-wise sigmoid activation:

$$h(z) = \frac{1}{1+e^{-\alpha(z-c)}}, \tag{2}$$

where $\alpha > 0$ is the stiffness parameter and $c > 0$ is a threshold.

The input is constrained to the box $0 \le x \le 1$, and the target is binary, $y \in \{0,1\}^N$. The reconstruction objective is

$$J(x) = \|y - f_\theta(x)\|_2^2. \tag{3}$$

The optimization problem is therefore

$$x^* = \arg\min_{0\le x\le 1} J(x). \tag{4}$$

This setting is simple enough to analyze carefully, yet expressive enough to exhibit non-convex behavior, gradient saturation, and sensitivity to both operator structure and optimization method.

## 4 Landscape Analysis

The objective function for the inverse reconstruction problem is defined as:

$$J(x) = \|y - h(Ax)\|_2^2. \tag{5}$$

The gradient is given by:

$$\nabla J(x) = 2A^T\left((h(Ax)-y)\odot h'(Ax)\right). \tag{6}$$

The derivative of the sigmoid is:

$$h'(z) = \alpha\, h(z)(1-h(z)). \tag{7}$$

We formalize the impact of the stiffness parameter $\alpha$ and the threshold $c$ on the gradient below.

[Gradient Bound] The magnitude of the gradient $\nabla J(x)$ is bounded by the stiffness $\alpha$. Specifically,

$$\|\nabla J(x)\|_2 \le \frac{\alpha}{2}\left\|A^T(h(Ax)-y)\right\|_2. \tag{8}$$

Since $h'(z) = \alpha h(z)(1-h(z))$, and the maximum value of $s(1-s)$ for $s\in[0,1]$ is $1/4$, we have $h'(z) \le \alpha/4$. Applying the triangle inequality to the gradient expression yields the result.

[Gradient Sparsity and Vanishing Gradients] For any fixed $x$ such that $(Ax)_i \ne c$ for all $i$, the gradient $\nabla J(x)$ vanishes as $\alpha\to\infty$:

$$\lim_{\alpha\to\infty} \nabla J(x) = 0. \tag{9}$$

As $\alpha\to\infty$, the sigmoid derivative $h'(z)$ approaches a Dirac delta function $\delta(z-c)$. If the pre-activation $(Ax)_i$ is not exactly equal to the threshold $c$, the derivative decays exponentially as $e^{-\alpha|(Ax)_i - c|}$. Consequently, for almost any $x$ not on the decision boundary, the gradient signal disappears.

As $\alpha$ increases, the landscape is transformed from a smooth, well-conditioned surface into one dominated by large flat plateaus separated by narrow transition boundaries. These boundaries act as geometric barriers.

Furthermore, the cutoff threshold $c$ dictates the centering of these informative regions. If $c$ is shifted away from the mean of $Ax$, the volume of the space with non-zero gradients shrinks further, forcing the optimizer to traverse larger regions of zero curvature where first-order information is absent.

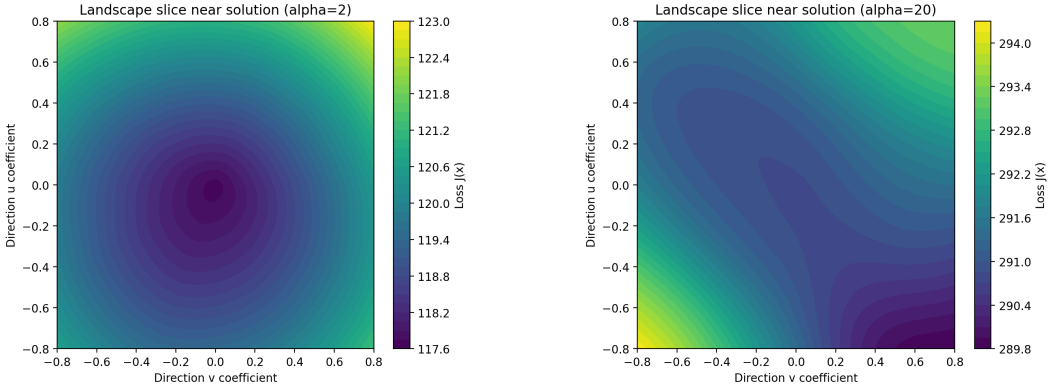


Figure 1: Two-dimensional landscape slices. Low $\alpha$ yields a smoother structure, while high $\alpha$ introduces distortion, sharper transitions, and flatter regions.

The convolution operator $A$ adds another layer of complexity by coupling neighboring variables. This spatial correlation leads to extreme anisotropy; the landscape becomes a series of curved, narrow valleys.

In high-$\alpha$ regimes, the gradient directions become unstable because small changes in $x$ can cause a coordinate to cross the threshold $c$, leading to a discontinuous change in the gradient signal.

Finally, we observe the property of non-uniqueness. Because of the null space of $A$ and the saturation regions of $h(\cdot)$, multiple distinct inputs $x$ can result in identical outputs $h(Ax)$. Geometrically, this implies that the landscape contains multiple global minima or flat valleys of equal depth.

In real-world inverse problems, this non-uniqueness suggests that achieving low loss $J(x)$ does not guarantee recovery of the true signal $x^*$, as the optimizer may converge to a valid but physically incorrect solution.

## 5 Algorithmic Analysis

Several optimization methods are considered in this study.

Projected Gradient Descent (PGD) relies directly on gradient magnitude and therefore struggles when gradients vanish. Momentum-based methods improve convergence by accumulating past updates but remain limited by gradient quality. Nesterov's accelerated gradient method extends this by applying a look-ahead correction before computing the gradient, yielding faster convergence in smooth regimes. However, this benefit diminishes in high-$\alpha$ regimes where the look-ahead step lands in an equally gradient-sparse region, offering little advantage over standard momentum.

Adaptive methods such as Adam rescale gradients coordinate-wise, allowing them to better handle ill-conditioning and sparse gradients. This makes them particularly effective when gradient magnitudes vary significantly across coordinates.

Quasi-Newton methods such as L-BFGS-B approximate curvature information and can accelerate convergence in smooth regimes. However, their performance degrades when gradients become sparse or unreliable.

The performance of these methods can be interpreted through the geometry described earlier. In low-$\alpha$ regimes, gradients are smooth and informative over a large region, so even basic methods can converge reliably. In high-$\alpha$ regimes, the problem becomes dominated by flat regions and sharply localized descent directions. Uniform step-size methods struggle because the few informative gradients are diluted by many inactive coordinates.

Adam mitigates this issue by adapting learning rates per coordinate based on historical gradient information. This allows it to preserve weak but consistent signals while reducing sensitivity to noisy directions. L-BFGS-B can perform very well when curvature information is stable, but it becomes less reliable when gradients are highly localized and the effective geometry changes abruptly.

Table 1: Qualitative comparison of the optimizers considered in this study.

| Optimizer | Robustness | Speed | Stability |
|---|---|---|---|
| PGD | Low | Slow | Low |
| Momentum | Medium | Medium | Medium |
| Nesterov | Medium | Medium | Medium |
| Adam | High | Fast | High |
| L-BFGS-B | Medium | Fast | Variable |

Table 1 summarizes the algorithmic behavior expected from the landscape analysis. These predictions are validated empirically in the next section.

# 6 Experiments

This section empirically validates the theoretical analysis of the optimization landscape. The goal is to understand how sigmoid stiffness, convolution structure, and target geometry affect optimization behavior, and how different algorithms respond to these changes.

All experiments are conducted on $64 \times 64$ inputs. Each configuration is evaluated over multiple random initializations, and results are reported using mean and standard deviation. Specifically, Adam uses $\text{lr} = 0.03$, $\beta_1 = 0.9$, $\beta_2 = 0.999$; PGD and Momentum use $\text{lr} = 0.1$ with momentum coefficient $\beta = 0.9$; Nesterov uses the same settings as Momentum; and L-BFGS-B uses its default line search. All methods run for 200 iterations over 5 random seeds.

## 6.1 Evaluation Metrics

Performance is evaluated using multiple complementary metrics:

- **Final loss** $J(x)$, measuring optimization accuracy.
- **Intersection-over-Union (IoU)**, measuring reconstruction quality in output space.
- **Relative loss reduction**, defined as

$$\frac{J(x_0) - J(x_T)}{J(x_0)}.$$

- **Success rate**, defined as the fraction of runs achieving IoU above a fixed threshold.

These metrics provide a comprehensive view of both optimization dynamics and reconstruction performance.

## 6.2 Effect of Sigmoid Stiffness

The sigmoid stiffness parameter $\alpha$ is the primary factor controlling landscape geometry.

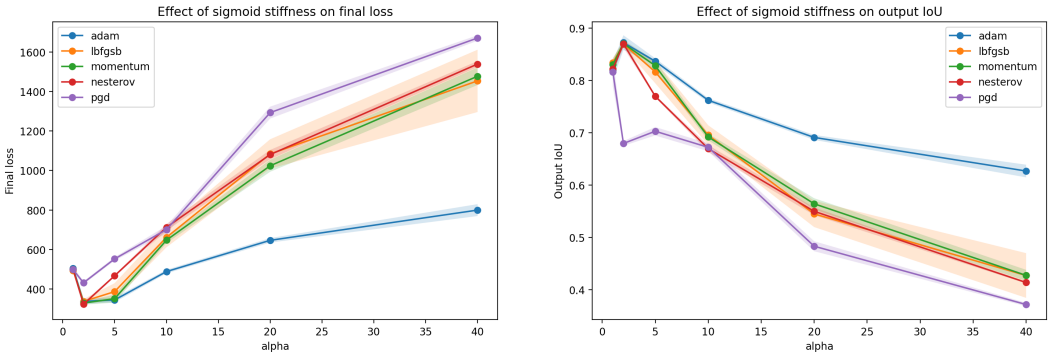


Figure 2: Effect of sigmoid stiffness on final loss and reconstruction IoU.

Performance degrades consistently as $\alpha$ increases. For small $\alpha$, the optimization landscape is smooth and well-conditioned, allowing all methods to perform effectively. As $\alpha$ increases, the landscape develops large flat regions and sharp transitions, making optimization significantly more difficult.

This behavior is explained by gradient sparsity: as $\alpha$ increases, the sigmoid derivative becomes concentrated near the threshold, reducing the fraction of active gradients.

Table 2: Performance across sigmoid stiffness values (Adam optimizer).

| $\alpha$ | Loss (mean $\pm$ std) | IoU (mean $\pm$ std) |
|---|---|---|
| 1.0 | $504.87 \pm 6.55$ | $0.8187 \pm 0.0174$ |
| 2.0 | $337.24 \pm 5.13$ | $0.8720 \pm 0.0140$ |
| 5.0 | $344.19 \pm 3.87$ | $0.8364 \pm 0.0056$ |
| 10.0 | $488.05 \pm 4.12$ | $0.7618 \pm 0.0041$ |
| 20.0 | $646.31 \pm 5.21$ | $0.6908 \pm 0.0047$ |
| 40.0 | $799.49 \pm 7.33$ | $0.6266 \pm 0.0121$ |

Loss increases substantially with $\alpha$, reflecting degraded optimization performance. Notably, IoU peaks at $\alpha = 2$ (0.872) rather than $\alpha = 1$ (0.819), and loss is non-monotone in the low-$\alpha$ regime reflecting that a slightly stiffer activation initially provides better conditioned gradients before sparsity dominates. From $\alpha = 2$ onward, IoU degrades monotonically and loss increases steadily, consistent with the theoretical prediction that gradient sparsity worsens optimization.

### 6.3 Impact of Cutoff Threshold $c$

While $\alpha$ controls the slope of the activation, the threshold $c$ determines the "centering" of the nonlinearity. I investigate how $c$ dictates the landscape geometry through a sensitivity sweep across $c \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

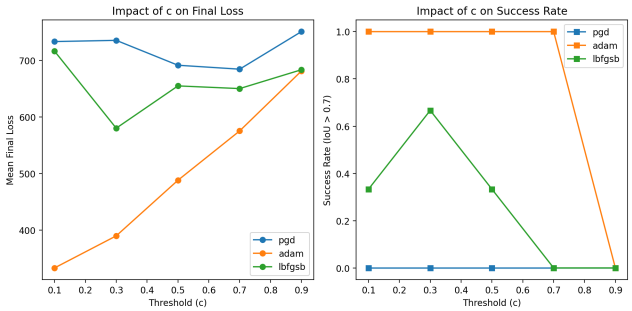


Figure 3: Sensitivity analysis of the cutoff threshold $c$. Left: Final optimization loss. Right: Success rate (IoU $> 0.7$).

The results in Figure 3 reveal a "V-shaped" performance profile. Optimization performance is maximized at $c = 0.5$ and degrades as the threshold approaches the boundaries of the input range.

This behavior is geometrically significant: when $c$ is shifted away from the mean of the pre-activation distribution $Ax$, a larger portion of the input space falls into the saturation regions of the sigmoid. This effectively "shifts" the informative part of the landscape away from the initialization, forcing the optimizer to traverse larger "flat" plateaus where the gradient $\nabla J(x)$ is nearly zero. Notably, Adam shows greater resilience to shifts in $c$ than PGD, as its adaptive scaling can amplify the vanishingly small gradients found in off-center threshold regimes.

### 6.4 Gradient Sparsity and Its Impact

To directly validate the theoretical explanation, the fraction of active gradients is measured.

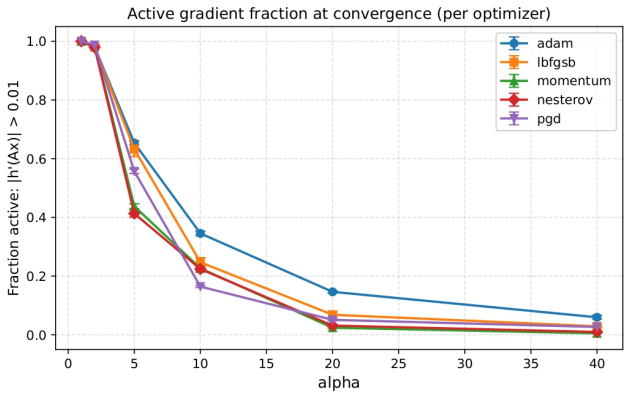


Figure 4: Fraction of active gradient components ($|h'(\mathbf{A}x)| > 0.01$) at convergence as of $\alpha$.

A sharp decline in gradient activity is observed as $\alpha$ increases for all optimizers, confirming that optimization difficulty arises from the collapse of gradient support. Critically, Adam maintains approximately $15\%$ active components at $\alpha = 20$, compared to under $5\%$ for PGD, Momentum, and Nesterov. This gap directly explains Adam's performance advantage: its coordinate-wise scaling allows it to navigate toward and remain in higher-gradient regions, while non-adaptive methods stagnate in sparser areas.

This has several consequences:

- Optimization effectively occurs in a lower-dimensional subspace.
- Gradient directions become unstable because small changes in $x$ can activate different coordinates.
- Convergence becomes slower and less reliable for non-adaptive methods.

These effects explain the degradation observed in both loss and IoU. The failure mode at large $\alpha$ is not simply "the problem gets a bit harder," but rather that the geometry changes qualitatively in a way that deprives the optimizer of useful information and adaptive methods are uniquely positioned to exploit whatever residual gradient signal remains.

### 6.5 Kernel Sensitivity

The convolution operator $A$ determines how variables interact, directly shaping the landscape.

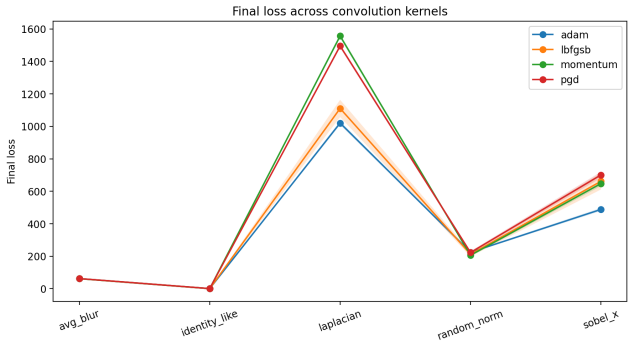


Figure 5: Final loss across convolution kernels for each optimizer. Identity-like kernels yield near-zero loss while Laplacian kernels produce the highest loss across all methods.

Identity-like kernels produce nearly separable problems, making optimization trivial. In contrast, Laplacian kernels introduce strong coupling between neighboring variables, resulting in complex curvature and higher difficulty, as quantified in Table 3.

Table 3: Kernel difficulty ranking based on average performance.

| Kernel | Difficulty |
|---|---|
| Identity-like | Very Easy |
| Average Blur | Easy |
| Random | Moderate |
| Sobel | Hard |
| Laplacian | Very Hard |

This demonstrates that landscape complexity is strongly influenced by operator structure. Problems that are nearly separable remain easy even with nonlinear activations, while strongly coupled operators create more challenging geometries. Importantly, optimization difficulty is not controlled by $\alpha$ alone. it also depends critically on how the forward operator mixes input coordinates.

### 6.6 Target Sensitivity

The structure of the target output also affects optimization.

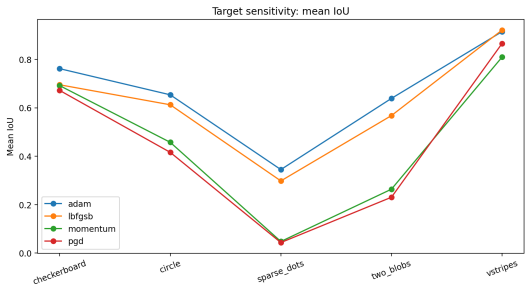


Figure 6: Reconstruction performance across different target patterns.

Structured targets such as stripes produce coherent gradient signals and are easier to reconstruct. Sparse or irregular targets generate fragmented gradients, making optimization more difficult.

This shows that the inverse problem depends not only on the model but also on the geometry of the desired output. In particular, targets that align with the structure of the convolution operator tend to be easier, while targets that require localized, irregular adjustments create more conflicting update directions.

### 6.7 Convergence Behavior

To analyze optimization dynamics, convergence curves are examined.

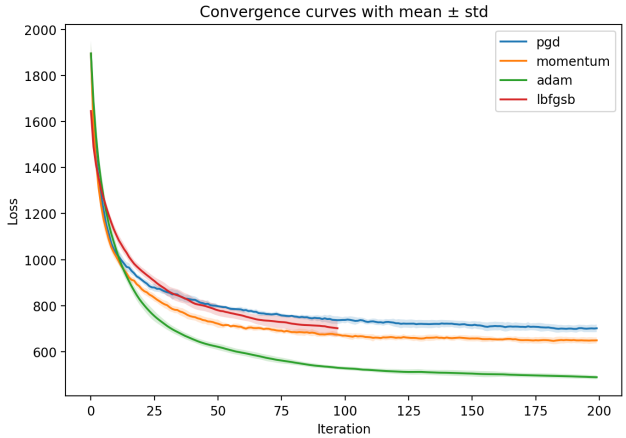


Figure 7: Convergence curves with mean and variance across runs.

Adam demonstrates both faster convergence and lower variance compared to other methods. PGD exhibits slower convergence and higher instability, particularly in high-$\alpha$ regimes.

This confirms that adaptive methods are better suited for landscapes with high anisotropy and sparse gradients. The gap between Adam and the non-adaptive methods is not only visible in final metrics, but also throughout the optimization trajectory.

### 6.8 Robustness and Variability

Variability across runs increases significantly with $\alpha$. In high-$\alpha$ regimes, small changes in initialization can lead to very different outcomes, indicating the presence of multiple local minima or attraction basins separated by flat regions.

Adam exhibits the lowest variance, while PGD and momentum methods show higher sensitivity to initialization. This observation is consistent with the landscape analysis: when gradients are sparse and localized, the algorithm can easily miss the narrow regions where useful descent information exists.

### 6.9 Detailed Interpretation of the Experimental Trends

While aggregate metrics such as final loss and IoU provide a high-level view of performance, a deeper understanding of optimization behavior requires examining how these metrics evolve across different regimes and configurations.

A consistent trend across all experiments is the progressive degradation of optimization performance as $\alpha$ increases. This degradation is not merely quantitative but reflects a qualitative shift in the nature of the optimization problem. For small values of $\alpha$, the objective behaves similarly to a smooth least-squares problem, where gradients are informative and well distributed across the domain. In this regime, most optimization algorithms perform comparably well.

However, as $\alpha$ increases, the sigmoid transitions toward a near-binary activation, effectively partitioning the space into regions of near-zero gradient separated by narrow transition bands. This leads to the emergence of large flat regions in the objective, where optimization algorithms receive little to no gradient signal. As a result, the optimization process becomes highly sensitive to initialization and step size.

The effective gradient can be written as

$$\nabla J(x) = 2A^T\big((h(Ax) - y) \odot h'(Ax)\big).$$

The term $h'(Ax)$ acts as a gate that determines which coordinates contribute to the update direction. When $\alpha$ is large, this gate becomes highly selective, allowing only a small subset of coordinates to influence the optimization. This reduces the effective dimensionality of the problem, introduces directional instability, and slows convergence.

### 6.10 Comparative Analysis Across Optimizers

A key goal of this study is to understand how different optimization algorithms respond to the landscape characteristics described above.

PGD performs adequately in low-$\alpha$ regimes but struggles as gradient sparsity increases. Because PGD applies uniform updates across all coordinates, it fails to exploit the limited regions where gradients are informative.

Momentum-based methods improve upon PGD by accumulating gradients over time, which helps smooth out noise and partially compensate for sparse updates. However, their performance still degrades in high-$\alpha$ regimes.

Adam consistently achieves the best performance across all configurations. Its adaptive learning-rate mechanism allows it to amplify weak but consistent gradients while suppressing noisy directions. This makes it particularly effective in navigating landscapes with highly uneven gradient distributions.

L-BFGS-B demonstrates fast convergence in smoother regimes but becomes less reliable as $\alpha$ increases. The curvature estimates it relies on become less informative when gradients are sparse and discontinuous.

### 6.11 Success Rate Analysis

To quantify robustness, success is defined as achieving IoU $> 0.7$. Figure 8 plots success rate versus $\alpha$ for each optimizer.

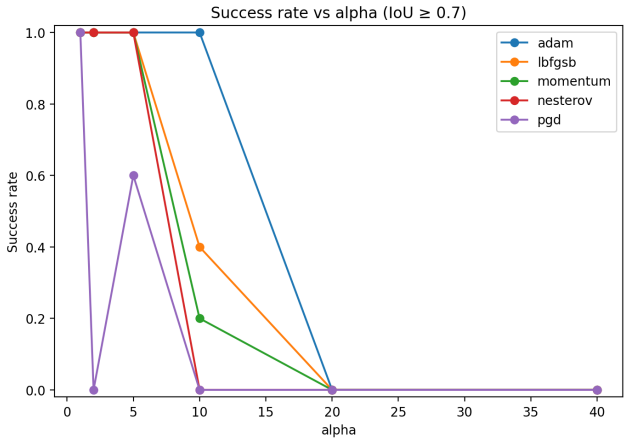


Figure 8: Success rate (IoU $> 0.7$) vs. sigmoid stiffness $\alpha$ across optimizers.

Success rates decrease sharply as $\alpha$ increases, particularly for non-adaptive methods. Notably, PGD shows a transient spike at $\alpha = 5$ attributable to favorable gradient alignment at that specific stiffness, before collapsing entirely at $\alpha = 10$. Adam maintains significantly higher success rates across all regimes, further confirming its robustness to sparse-gradient landscapes.

### 6.12 Qualitative Analysis and Non-Uniqueness

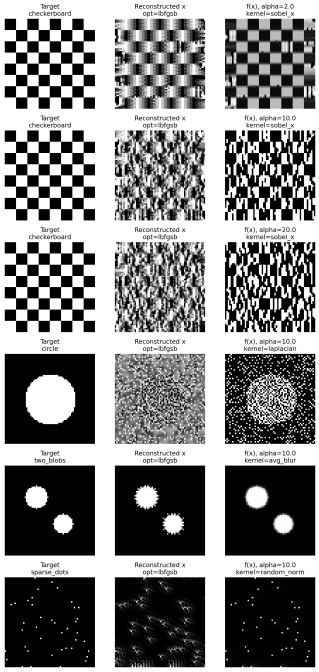


Figure 9: Representative reconstruction results.

Qualitative results reveal that reconstructed inputs often differ from the target structure, even when output accuracy is high. This demonstrates the non-uniqueness of the inverse problem.

Multiple inputs can produce similar outputs, which explains the presence of multiple local minima and flat regions in the landscape. This is an important point: success in output space does not imply recovery of a unique input. Instead, the optimizer may converge to one of several valid solutions, depending on initialization and algorithmic behavior.

### 6.13 Discussion

Three cross-cutting observations emerge from the experiments above. First, increasing $\alpha$ induces an *implicit dimensional collapse*: as the sigmoid steepens, only a small subset of coordinates retains non-zero gradient information (Figure 4), forcing optimization into an increasingly low-dimensional subspace. Second, high-$\alpha$ regimes produce characteristic *failure modes* the optimizer stalls in flat plateaus where no descent direction is available, and sensitivity to initialization grows sharply, as evidenced by the rising variance in Figure 7. Third, the gradient gate $h'(Ax)$ is active only when $Ax \approx c$, corresponding to edge-like transition regions in the target. This produces *edge-driven dynamics*: early iterations reconstruct coarse boundaries while later iterations fill interior regions — a direct consequence of the nonlinearity's spatial selectivity.

## 7 Conclusion

This work reveals that sigmoid stiffness fundamentally reshapes the optimization landscape by inducing gradient sparsity and effective dimensional collapse. In this regime, reconstruction becomes an edge-driven process, where information propagates from boundary regions inward.

Our findings highlight the importance of adaptive optimization methods in navigating such landscapes and suggest that even simple architectures can exhibit highly non-trivial optimization behavior.

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